# Chapter 10

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## Problem 16

We are considering some system $S$ satisfying conditions $L_1, L_2$ with any disjoint number sets $A, B$. In addition, we assume that $A$ is enumerated by the formula $F_1(x, y)$, while $B$ is enumerated by the formula $F_2(x, y)$. We are trying to show that $B$ is strongly separated from $A$ by the formula $F(x)$:
$\forall y[F_1(x, y) \supset \exists z(z \leq y \wedge F_2(x, z))]$.
Thus, we must show the following holds for some superset $S_1$ of $A$, and some superset $S_2$ of $B$:
$C_1$ : If $n \in S_2$, then $F(\overline{n})$ is provable.
$C_2$ : If $F(\overline{n})$ is provable, then $n \in S_2$.
$C_3$ : If $n \notin S_2$, then $\sim F(\overline{n})$ is provable.
$C_4$ : If $\sim F(\overline{n})$ is provable, then $n \notin S_2$.
Define $N_1$ and $N_2$ to be the sets of all $n$ such that $F(\overline{n})$ and $\sim F(\overline{n})$ are provable, respectively. Now, define $S_1$ to be $A \bigcup N_1$ and $S_2$ to be $B \bigcup N_2$. Because of the assumed simple consistency and the fact that $A$ and $B$ are disjoint, $S_1$ and $S_2$ are similarly disjoint.
Proof for $C_1$:

*Proof.* There are two cases: $n \notin B$ or $n \in B$. In the first case, $F(\overline{n})$ was defined to be provable for this $n$. Otherwise, we have $\overline{n} \in B$. Thus, $F_1(\overline{n}, y)$ is refutable for every $y$, and thus, by $L_1$, $\forall y(y \leq \overline{n} \supset \sim F_1(\overline{n}, y))$ is provable. Also, because $\overline{n} \in S_2$, $F_2(\overline{n}, m)$ is provable for some $m$. Hence, $\forall y[\overline{k} \leq y \supset \exists z(z \leq y \wedge F_2(\overline{n}, z))]$ (because $\overline{k} \leq y$, $F_2(\overline{n}, z)$ is provable when $z = k$). Thus, by $L_2'$, $\forall y[\sim F_1(\overline{n}, y) \vee \exists z(z \leq y \wedge F_2(\overline{n}, z))]$ is provable, meaning the equivalent formula $\forall y[F_1(\overline{n}, y) \supset \exists z(z \leq y \wedge F_2(\overline{n}, z))]$ is also provable. □

Proof for $C_2$:

*Proof.* Temporarily assume that $n \notin S_2$. This implies that it is neither in $B$ or some $n$ for which $F(\overline{n})$ is provable, thus yeilding a contradiction. □

Proof for $C_3$:

*Proof.* There are two cases: $n \notin A$ or $n \in A$. In the first case, $\sim F(\overline{n})$ was defined to be provable for this $n$. Otherwise, we have $\overline{n} \in A$. Thus, $F_1(\overline{n}, y)$ is provable for some $m$. Also, because $n \notin B$, $F_2(\overline{n}, y)$ is refutable for every $m$, and hence, by $L_1$, we have:
(1) $\forall z(z \leq \overline{m} \supset \sim F_2(\overline{n}, z))$
Hence, we have:
(2) $F_1(\overline{n}, \overline{m}) \wedge \forall z(z \leq \overline{m} \supset \sim F_2(\overline{n}, z))$
Then, it can be seen that (1) implies:
(3) $\sim \exists z(z \leq \overline{m} \wedge F_2(\overline{n}, z))$
Thus, substituting (3) for (1) in (2) yields:
(4) $F_1(\overline{n}, \overline{m}) \wedge \sim \exists z(z \leq \overline{m} \wedge F_2(\overline{n}, z))$
which implies:
$\sim[F_1(\overline{n}, \overline{m}) \supset \exists z(z \leq \overline{m} \wedge F_2(\overline{n}, z))]$
which implies:
$\exists y \sim[F_1(\overline{n}, y) \supset \exists z(z \leq y \wedge F_2(\overline{n}, z))]$
which can be equivalently stated as: $\sim \forall y[F_1(\overline{n}, y) \supset \exists z(z \leq \overline{m} \wedge F_2(\overline{n}, z))]$
□

Proof for $C_4$:

*Proof.* Temporarily assume that $n \notin S_1$. Hence it would not be one of those $n$ for which $F(\overline{n})$ is provable, thus yeilding a contradiction. □

# Problem 17

We are considering some system $S$ obeying the Rosser conditions that is simply consistent. We are to show that there is an undecidable sentence.

*Proof.* Because we are given that both $P^*, R^*$ are enumerable, we know that $R^*$ is strongly separable from $P^*$ by the Separation Lemma for Sets. Thus, by problem 8(b), we know that $R^*$ is also weakly separable from $P^*$. Hence, by problem 7(a), there is an undecidable sentence. In particular, we know that, if $F_1(x, y), F_2(x, y)$ enumerate $P^*, R^*$, respectively, the formula $F(x)$:
$\forall y[F_1(x, y) \supset \exists z(z \leq y \wedge F_2(x, y))]$
strongly separates $R^*$ from $P^*$. Then, if $f$ is the Gödel number for $F(x)$, problem 7(a) shows that $F(\overline{f})$ is an undecidable sentence. □

## Exercise

We are considering some system $S$ obeying $L_1, L_2$ in which $R_1, R_2$ are two $n$-ary relations enumerated by $F_1(x_1, ..., x_n, y)$ and $F_2(x_1, ..., x_n, y)$, respectively. We are to show that $R_2 - R_1$ is strongly separated from $R_1 - R_2$ by the formula $F(x)$:
$\forall y[F_1(x_1, ..., x_n, y) \supset \exists z(z \leq y \wedge F_2(x_1, ..., x_n, z))]$

*Proof.* In the first case, assume that $R_2 - R_1$ holds. Hence, $R_2$ holds, while $R_1$ does not hold. Thus, for some $m$, $F_2(\overline{x}_1, ..., \overline{x}_n, \overline{m})$ holds, thus implying that:
$\forall y[\overline{m} \leq y \supset \exists z(z \leq y \wedge F_2(\overline{x}_1, ..., \overline{x}_2, z))]$ is provable. Also, $F_1(\overline{x}_1, ..., \overline{x}_n, \overline{m})$ is refutable for every $m$, thus implying that:
$\forall y[y \leq \overline{m} \supset \sim F_1(\overline{x}_1, ..., \overline{x}_n, \overline{m}))]$ is provable by $L_1$. Hence, by $L_2'$, we have:
$\forall y[\sim F_1(\overline{x}_1, ..., \overline{x}_n, \overline{m}) \vee \exists z(z \leq y \wedge F_2(\overline{x}_1, ..., \overline{x}_2, z))]$ which implies:
$\forall y[F_1(\overline{x}_1, ..., \overline{x}_2, \overline{m}) \supset \exists z(z \leq y \wedge F_2(\overline{x}_1, ..., \overline{x}_2, z))]$

In the second case, assume that $R_1 - R - 2$ holds. Hence, $R_1$ holds, while $R_2$ does not hold. Thus, $F_1(\overline{x}_1, ..., \overline{x}_n, \overline{m})$ is provable for some $m$, while $F_2(\overline{x}_1, ..., \overline{x}_n, \overline{m})$ is refutable for every $m$. The rest of this case is identical to that in the proof for the Separation Lemma for Relations. □